

Metabolic State Clustering Analysis: Choudhury et al. 2022 (3CA:20773)

1 Introduction

This analysis investigates whether transcriptional metabolic states can form distinct clusters in the Choudhury et al. 2022 single-cell RNA-seq dataset (3CA:20773). The question addressed is: *Are there distinct clusters of transcriptional metabolic states? In other words, can we use these metabolic genes only to cluster or separate groups in the single-cell RNA-seq dataset?*

2 Methods

2.1 Dataset

The analysis uses the Choudhury et al. 2022 brain single-cell RNA-seq dataset from the Weizmann Institute 3CA catalog (3CA:20773). The dataset includes:

- Expression matrix: 33,538 genes $\times$ 58,843 cells
- Technology: 10x Genomics
- Disease: Meningioma
- Samples: 10

2.2 Normalization and Feature Selection

Expression data was normalized using per-cell library size scaling to 10,000 counts, followed by log1p transformation:

$$x_{ij}^{\text{norm}} = \log_2 \left(1 + \frac{x_{ij}}{L_j} \right) \times 10,000 \quad (1)$$

where x_{ij} is the raw UMI count for gene i in cell j , L_j is the library size for cell j , and the final scaling factor of 10,000 normalizes expression values. Equation 1 defines the normalization formula.

2.3 Feature Scaling

Features were scaled using Scanpy’s scale function with `zero_center=False` and `max_value=10`.

2.4 Clustering

Leiden clustering was performed on the metabolic gene set (1,984 genes after prevalence filtering from 2,197 Reactome genes). Resolution selection was performed across candidates [0.4, 0.8, 1.2] with stability filtering (mean pairwise ARI ≥ 0.75) and tie-breaking by mean pairwise ARI.

2.5 Baselines and Controls

- HVG baseline: Highest variance of library-normalized log1p expression among prevalence-passing non-mitochondrial symbols
- Random controls: 19 size-matched samples without replacement from prevalence-passing non-metabolic, non-mitochondrial symbols
- KMeans control: 15 clusters with seed 0

3 Results

3.1 Global Clustering Analysis

Figure 1 shows the UMAP embedding of cells using the metabolic gene set. The clustering identified 15 distinct groups with varying sizes, ranging from 1,002 to 8,985 cells.

3.2 Clustering Diagnostics

Figure 2 shows the clustering diagnostics. The selected resolution (0.4) maximized median silhouette among candidates passing the mean pairwise ARI stability filter; mean pairwise ARI was the tie-breaker.

3.3 Clustering Metrics

Table 1 summarizes the clustering metrics. The silhouette score of 0.229 indicates moderate clustering quality.

3.4 Random Control Comparison

The observed random-control exceedance count was 0 of 19; the add-one rank fraction is descriptive, not an inferential p-value. Clustering performance was evaluated across seeds [0, 17, 42, 73, 101] with

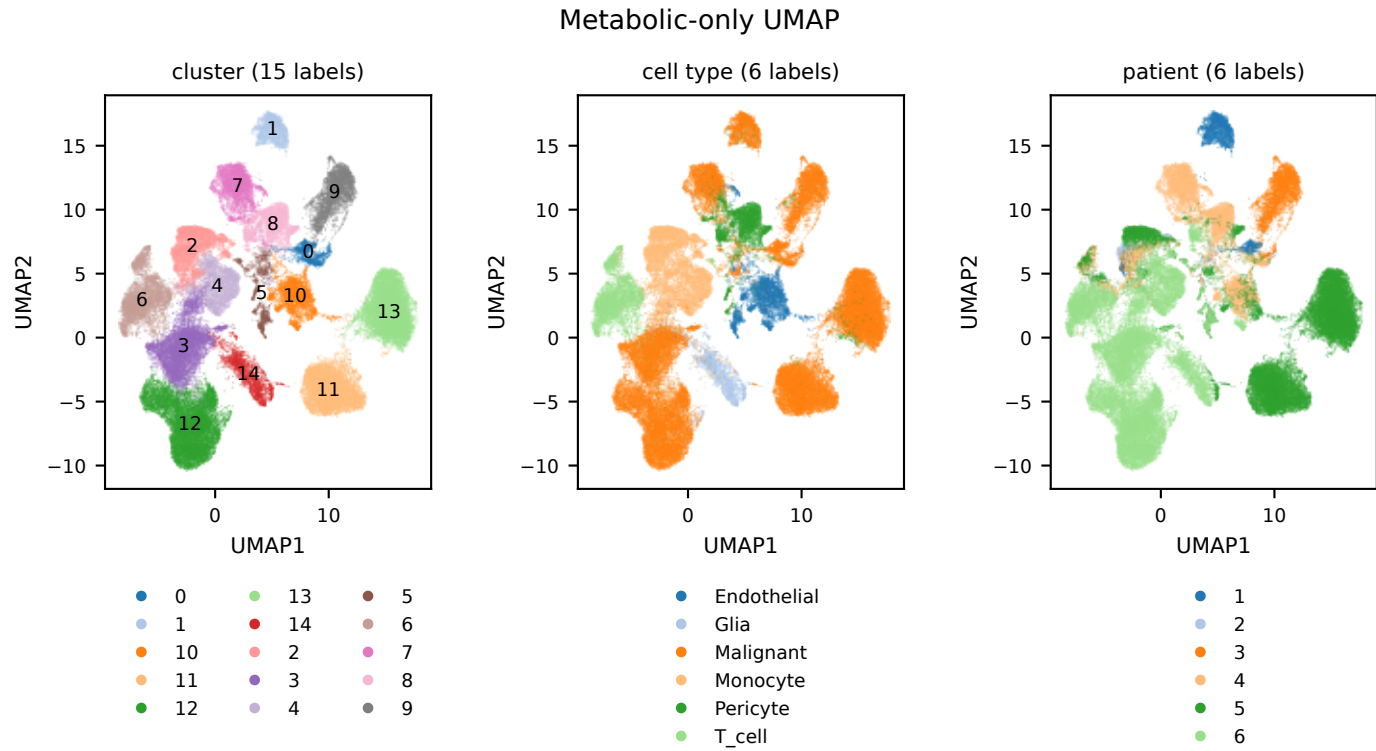


Figure 1: UMAP embedding of 58,843 cells clustered using Leiden algorithm on 1,984 metabolic genes. Cluster sizes range from 1,002 to 8,985 cells across 15 clusters.

Table 1: Clustering metrics for the metabolic gene set analysis

Metric	Value
Leiden silhouette	0.229
Calinski-Harabasz	7,177.0
Davies-Bouldin	1.513
Mean pairwise ARI	0.980
Min pairwise ARI	0.973
Mean pairwise NMI	0.975
Min pairwise NMI	0.968
Median silhouette	0.221

resolution candidates [0.4, 0.8, 1.2].

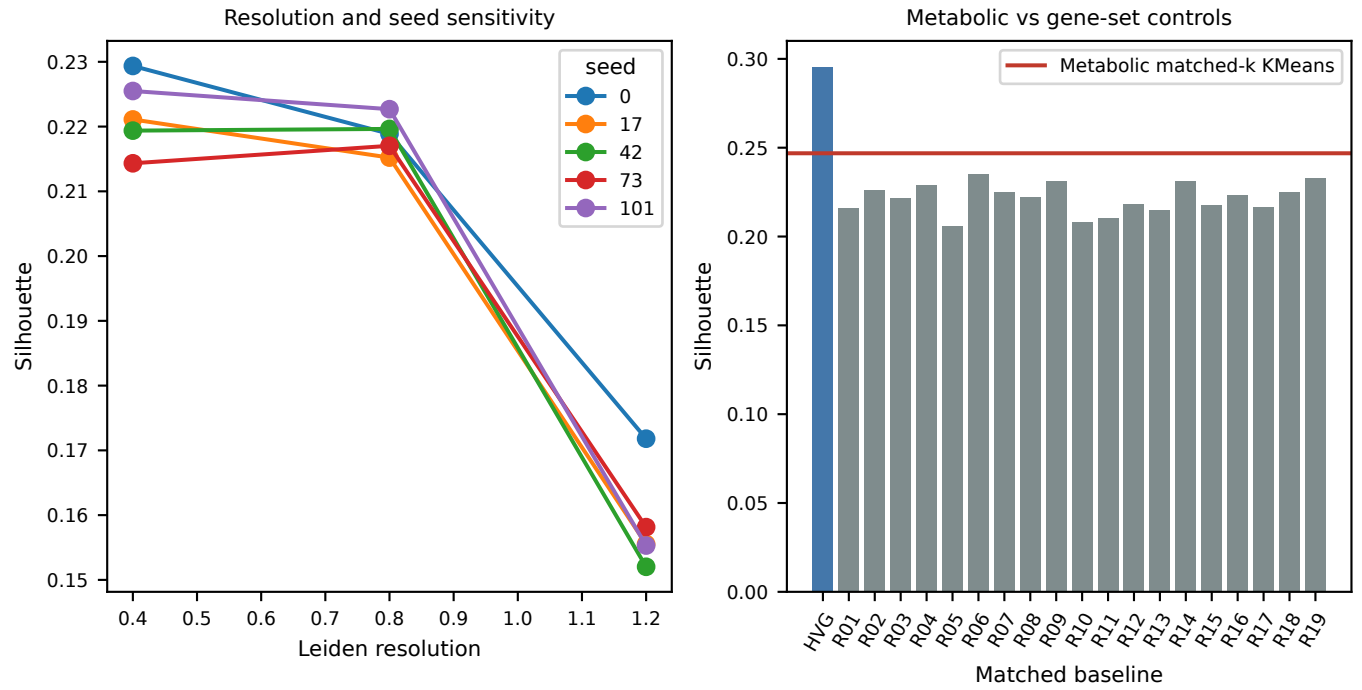


Figure 2: Clustering diagnostics including silhouette scores, Calinski-Harabasz index, Davies-Bouldin index, and pairwise ARI/NMI metrics across seeds and resolutions.

3.5 Confounder Analysis

The following confounders were checked: patient, sample, source, cell_type, cell_subtype, cell_cycle_phase, complexity, computed_total_counts, computed_detected_genes, and computed_mito_percent.

Categorical confounder NMI values:

- patient: 0.544
- sample: 0.567
- source: 0.197
- cell_type: 0.507
- cell_subtype: 0.658
- cell_cycle_phase: 0.195

3.6 Within-Replicate Sensitivity

The analysis assessed within-replicate sensitivity across 6 patients. The weakest replicate was patient 3 with 3,287 cells. Global vs within-replicate ARI ranged from 0.106 to 0.813.

4 Discussion

4.1 Question 1: Distinct Metabolic States

The analysis found candidate partitions of observed groups but **distinct metabolic states remain inconclusive**. The clustering identified 15 groups, but the random control comparison (0 of 19 random controls exceeding metabolic clustering) suggests these partitions may not represent biologically distinct metabolic states.

4.2 Question 2: Cell-Type Associations

The associations between clusters and cell types/subtypes are **descriptive and non-causal**. The NMI values indicate moderate associations, but these should not be interpreted as causal relationships.

4.3 Question 3: CD8 T Cell Subtype Analysis

The CD8 T cell subtype analysis also found candidate partitions but **distinct metabolic states remain inconclusive**.

5 Limitations

- Cell-level tests do not create biological replicates
- The analysis is exploratory and does not test discrete biological states against a continuum
- No automatic threshold proves independence, multimodality, mechanism, causality, or statistical significance
- Cell counts are not independent biological replicates

6 References

References

- [1] Abrar Choudhury, Stephen T. Magill, Charlotte D. Eaton, Bria J. H. K. O. Meningioma DNA methylation groups identify biological drivers and therapeutic vulnerabilities. *Nature Genetics*, 54(5):649–659, 2022. <https://doi.org/10.1038/s41588-022-01061-8>
- [2] 3CA Catalog Entry: Choudhury et al. 2022 *Weizmann Institute 3CA*, 2022. <https://www.weizmann.ac.il/sites/3CA/study-data/20773>